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Session : Conference Thematic

REVOLUTIONIZING ENERGYMANAGEMENT WITH MATTER PROTOCOL

Presented By RENAULT

Nirmala Mantri, Deputy Manager, Renault Nissan Technology & Business Centre India

INTRODUCTION

Electric Vehicle Adoption

Growing use of electric vehicles increases the need for integrated home management.

30%

EV Market Share

Projected electric vehicle sales by 2030

Limitations of Traditional Systems

Isolated energy systems create inefficiencies, limiting real-time optimization and demand response.

Matter Protocol Integration

Matter protocol enables unified communication between smart meters, EV chargers, and appliances.

Smart Home Energy Management

SHEMS empower homes to contribute to grid stability through advanced energy control.

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CONTEXT

Smart Home Energy Systems

Smart Home Energy Systems enable monitoring and optimization of residential energy consumption for flexible demand management.

Interoperability Challenges

Fragmented ecosystems like HomeKit, Google Home, and Alexa limit unified automation and large-scale energy management adoption.

Matter Protocol Benefits

Matter protocol bridges heterogeneous IoT devices with IP-based, encrypted, cloud-independent communication for smart grids.

Expanding Matter's Applications

Applying Matter to smart meters and EV chargers enhances energy infrastructure resilience and security.

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RELEVANCE

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Interoperable Energy Solutions

Matter protocol enables seamless communication across different device manufacturers and home automation systems.

Energy Efficiency and Demand Flexibility

Seamless device communication encourages energy savings and enhances participation in demand-side management programs.

Modern IP-Based Communication

Matter supports Wi-Fi and Thread protocols, ensuring high reliability and speed for devices like EV chargers and smart meters.

Alignment with Smart Grid Goals

Interoperable technology supports data-driven utilities and dynamic pricing, empowering customers to respond to grid signals.

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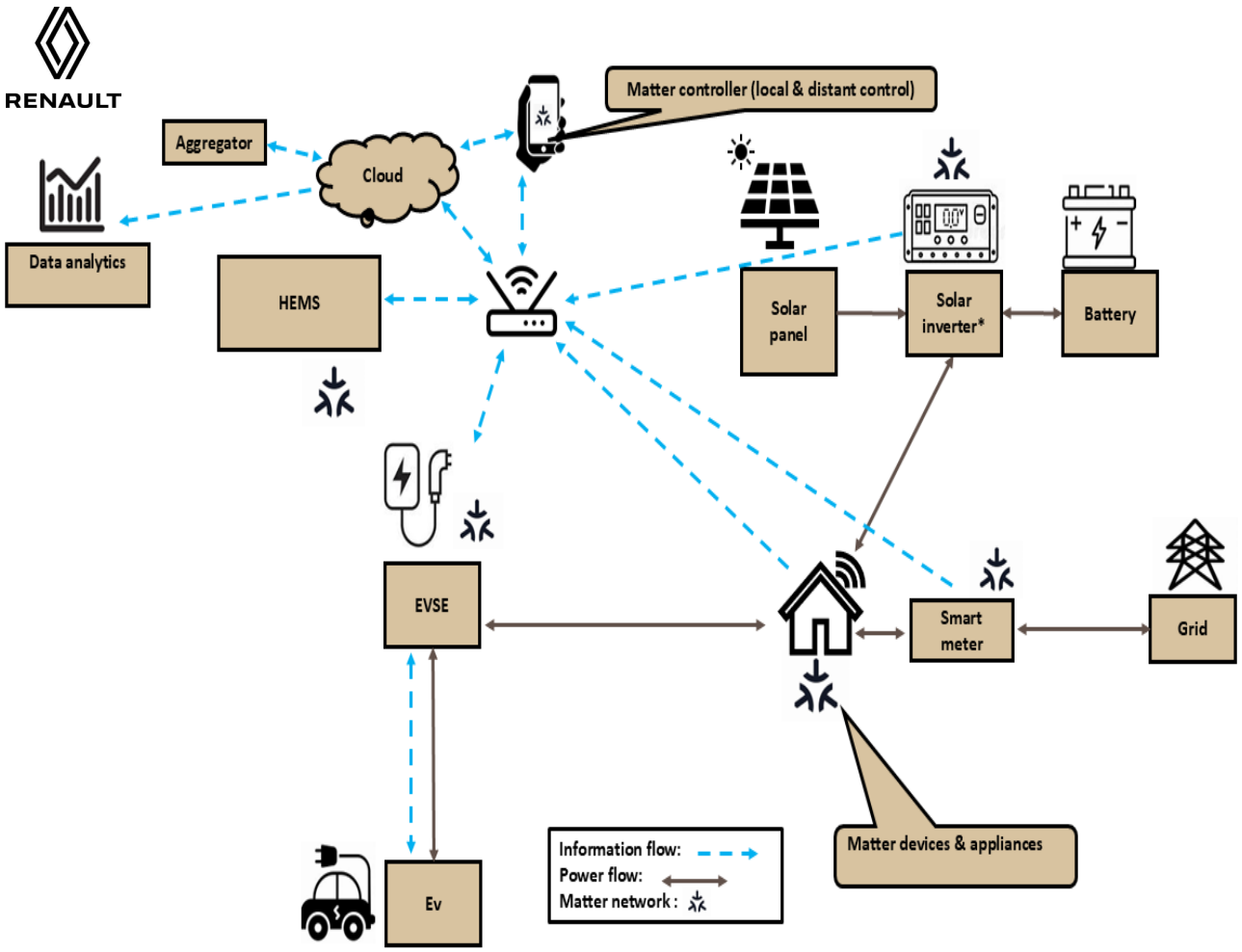
Matter-Enabled Smart Home Energy Framework



- Integrates smart meter, EV charger, solar inverter, and battery storage.
- Matter controller enables bi-directional control and real-time energy data exchange.
- Supports local + remote automation through IPv6-based communication.
- Improves home-to-grid coordination and energy optimization

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METHODOLOGY

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EV Charger Integration

EV charger → MCU-based Matter bridge → Amazon Echo Dot (Matter controller)

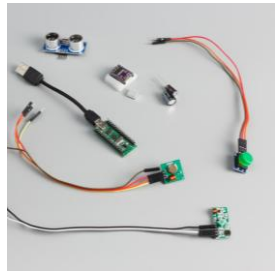
Enables EV charger monitoring + On/Off control through Matter clusters

Smart Meter Integration

Smart meter data read via Matter-enabled reader

Supports visualization in Home Assistant & cloud platforms

Facilitates remote energy monitoring and analytics



EV Charger Communication

A microcontroller-based Matter bridge converts HTTP EV charger communication into Matter-compliant clusters for smart control.

Smart Meter Data Transmission

Smart meter integrated with Raspberry Pi and Home Assistant enables live consumption data visualization via Matter-enabled readers.

Cloud Integration and Monitoring

Cloud integration using REST services demonstrates Matter's support for remote monitoring of energy systems.

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KEY RESULTS

Matter Protocol Validation

The case study validates Matter protocol managing EV charger control and smart meter data seamlessly over Wi-Fi.

Data Collection and Command Execution

EV charger logs confirmed command execution; smart meter logs showed voltage, current, and power data ingestion.

Integration and Future Potential

Mapping EV charger APIs and smart meter telemetry to Matter clusters proves its viability for smart energy management.

CONCLUSION

Unified Energy Control

Matter enables bridging of smart meters, EV chargers, and controllers into one intelligent ecosystem for energy management.

Scalable Smart Home Framework

The protocol supports consistent command execution, data visibility, and real-time control for scalable energy systems.

Future Opportunities

Matter's potential for advanced analytics, smart grid integration, and expanded energy resource applications promises a sustainable future.

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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

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Links/References (If any)